Springboard virtual Internship 6.0

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Collage : KL University

Branch : CSE

Domin :Data Visualization

Project : Visualizing US Natural Disaster Declarations

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Milestone-3 Documentation: Geospatial Disaster Analysis and Visualization

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1. Objective

The objective of this milestone is to analyze the geographical distribution of natural disaster declarations across the United States.
The goal is to understand how disasters vary across different states and identify regions that experience disasters more frequently.
By converting tabular data into maps and spatial visualizations, hidden patterns and geographic hotspots can be identified.
This milestone focuses on geospatial analysis and visualization techniques to better understand disaster distribution.

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2. Problem Formulation

Analysis Type:
- Geospatial Data Analysis
- Exploratory Data Analysis (EDA)

Key Questions:
1. Which states have the highest number of disaster declarations?
2. Are disasters concentrated in specific regions of the United States?
3. Do certain types of disasters occur more frequently in specific states?
4. Can disaster hotspots be identified using spatial visualizations?

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3. Dataset Preparation for Geographic Analysis

Input Dataset:
usnd_cleaned.csv (cleaned dataset generated during Milestone-1)

Required Columns:
- state
- incidentType

- year
- month

Data Validation Steps:
- Ensure state codes or names are consistent.
- Check for missing state values.
- Verify dataset contains records across multiple states.
- Confirm there are no invalid or null values.

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4. Geographic Aggregation

Geographic aggregation groups disaster records by location.

4.1 State-Level Aggregation:
The dataset is grouped by state to calculate total disaster declarations per state.
Purpose:
- Identify states with highest disaster occurrences.
- Understand spatial distribution of disasters.

4.2 State + Incident Type Aggregation:
Group the dataset by state and incident type.
Purpose:
- Identify dominant disaster types in each state.

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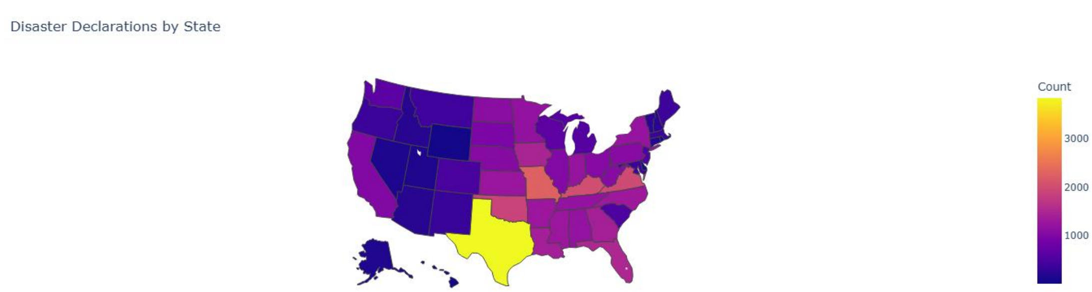
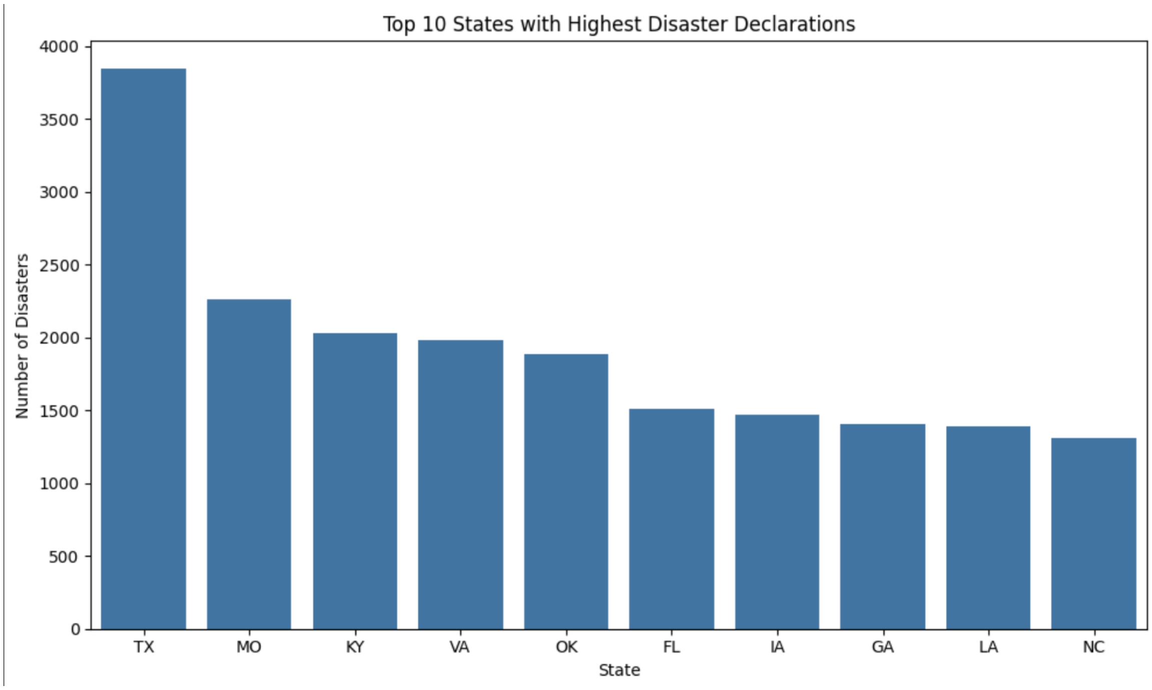
5. Core Visualizations

5.1 Choropleth Map
Displays disaster frequency across states using color intensity.
Darker colors represent higher disaster counts.



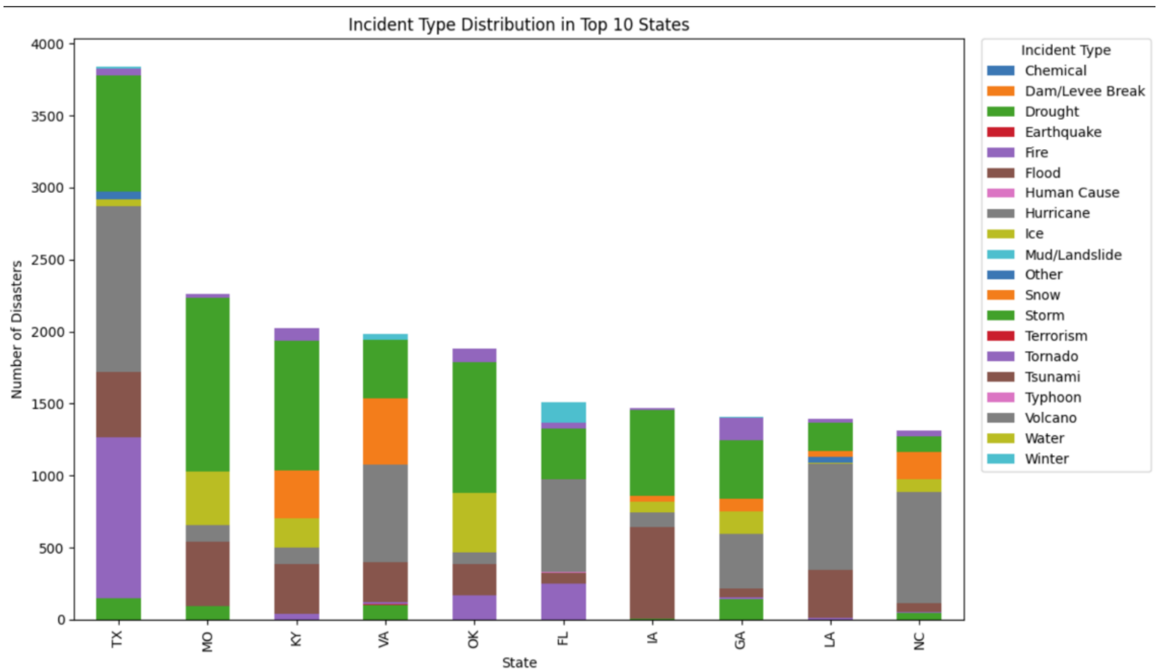
5.2 State Comparison Chart

A bar chart comparing disaster counts across the top 10 states.



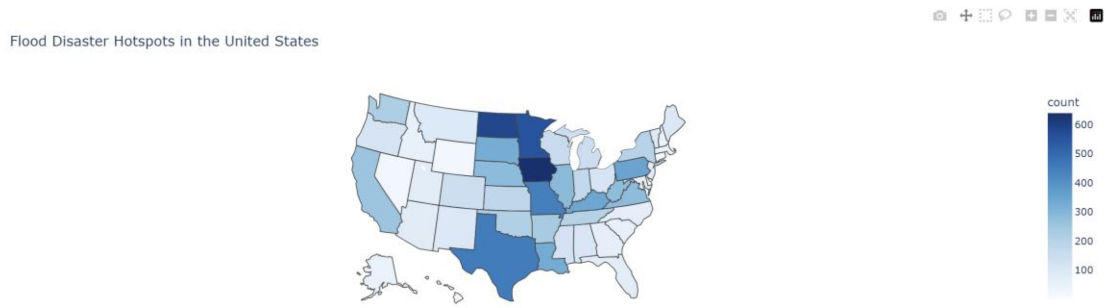
5.3 Incident Type Distribution by State

Stacked bar chart showing how different disaster types contribute to total declarations in each state.



5.4 Incident Type Hotspot Map

Maps highlighting hotspots for specific disaster types such as hurricanes, floods, or fires.



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6. Tools and Libraries Used

Data Processing:
- Pandas
- NumPy

Visualization:
- Matplotlib
- Seaborn
- Plotly

Geospatial Analysis:
- GeoPandas
- Folium

These libraries help process data and create both static and interactive maps.

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7. Visualization Standards

Mandatory Elements:
- Title
- Axis Labels
- Legend
- Proper Color Scale

Best Practices:
- Use readable color schemes
- Avoid overcrowded maps
- Highlight key insights clearly
- Ensure charts are easy to interpret

9.Conclusion

The Geospatial Disaster Analysis successfully visualized the geographical distribution of disaster declarations across the United States. By aggregating the cleaned dataset by state and incident type, meaningful patterns in disaster occurrences were identified.

The visualizations such as choropleth maps, bar charts, and stacked charts helped highlight disaster-prone regions and showed how different types of disasters vary across states. Some states were observed to have higher disaster frequencies due to geographic and climatic conditions.